

EE 101 ANALOG ELECTRONICS

- Diode, Zener Diodes, Rectifier Circuits, Wave Shaping Circuits (3 Hours)
- Bipolar Junction Transistors, Transistor Biasing (2.5 hours)
- Transistor Small Signal Analysis, Transistor Amplifiers. (2.5 hours)
- Op-amp Circuits (2 hours)

Textbooks: Analog Electronics

1. R.L. Boylestad and L. Nashelsky: Electronic Devices and Circuit Theory; Pearson Education.
2. A.S. Sedra and K.C. Smith: Microelectronic Circuits: Theory and Applications, Oxford University Press

Linear Elements in a Circuit

Elements Symbol	RESISTOR 	CAPACITOR 	INDUCTOR 
Denoted by	R	C	L
Equation	$R = \frac{V}{I}$	$C = \frac{Q}{V}$	$L = \frac{V_L}{(di/dt)}$
Series	$R_T = R_1 + R_2$	$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$	$L_T = L_1 + L_2$
Parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	$C_T = C_1 + C_2$	$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2}$

Topic - DIODES

Non linear elements that we will study



Resistor



Capacitor

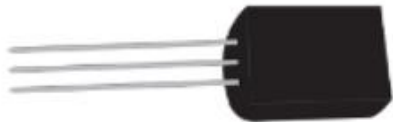


Inductor



Diode

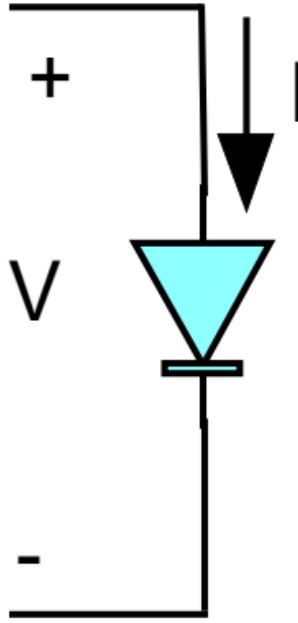
- Two terminal Electronic component with Non linear behaviour in V -I characteristics



Transistor

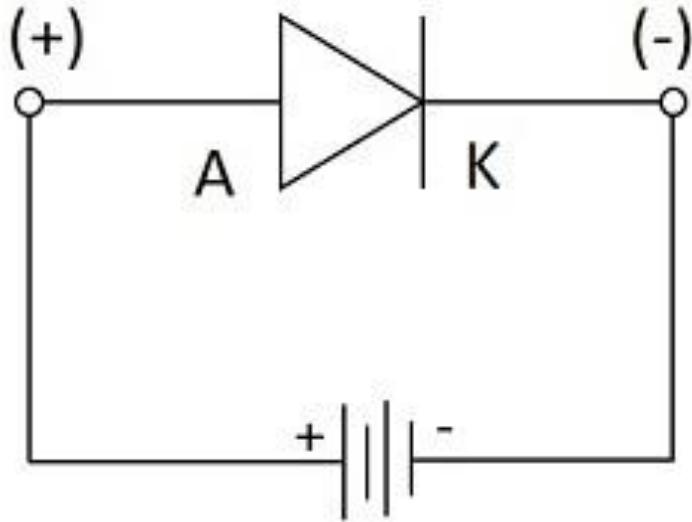
- Three terminal Electronic component with Non linear behaviour in V -I characteristics

Diodes

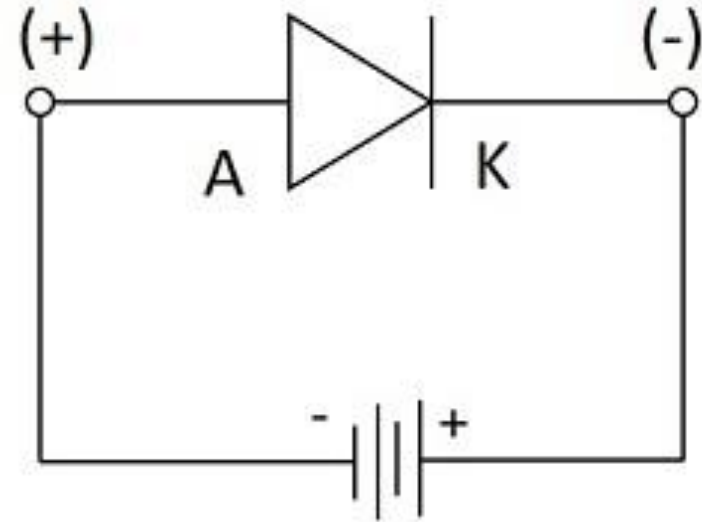


Diode Properties

- Two-terminal device that conducts current freely in one direction but blocks current flow in the opposite direction.
- The two electrodes are the anode which must be connected to a positive voltage with respect to the other terminal, the cathode in order for current to flow.



Forward biased Connection

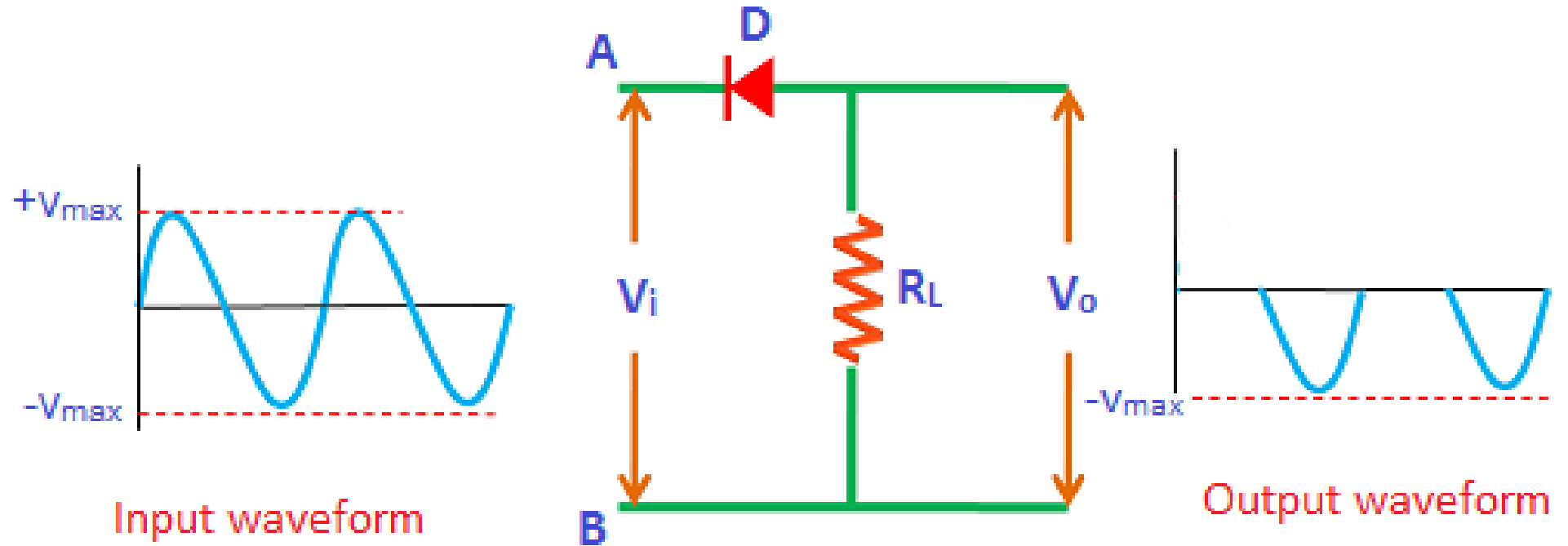


Reverse biased Connection

A diode conducts well in forward biased condition.

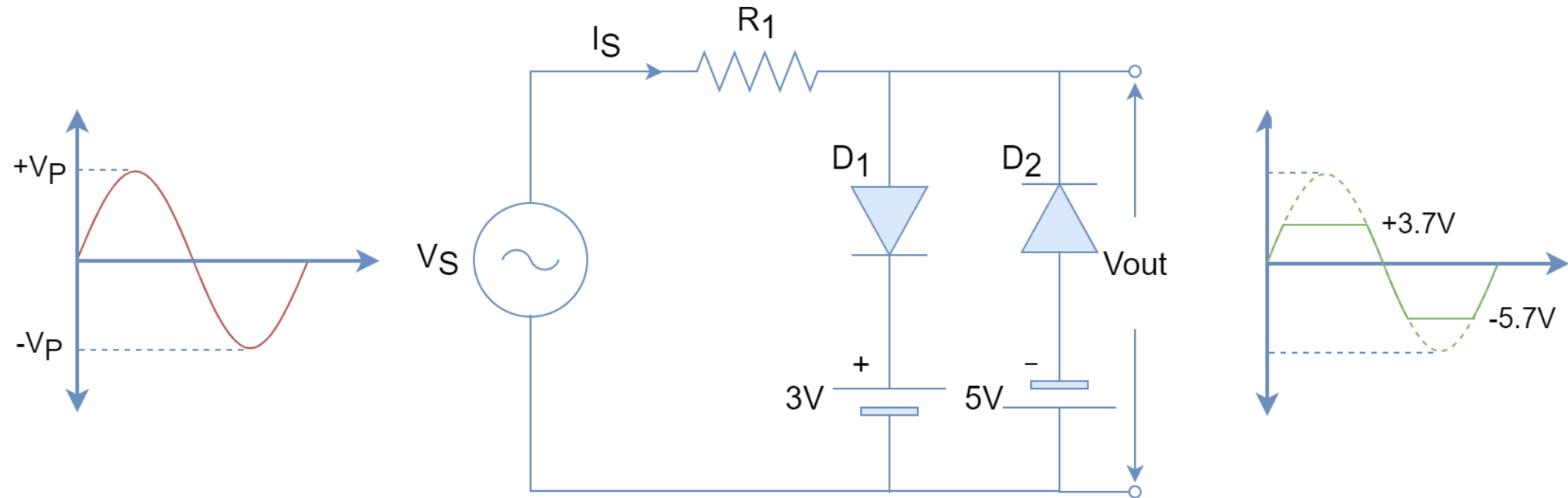
A diode cannot conduct in reverse biased condition.

Application of Diode circuits (Clipping)

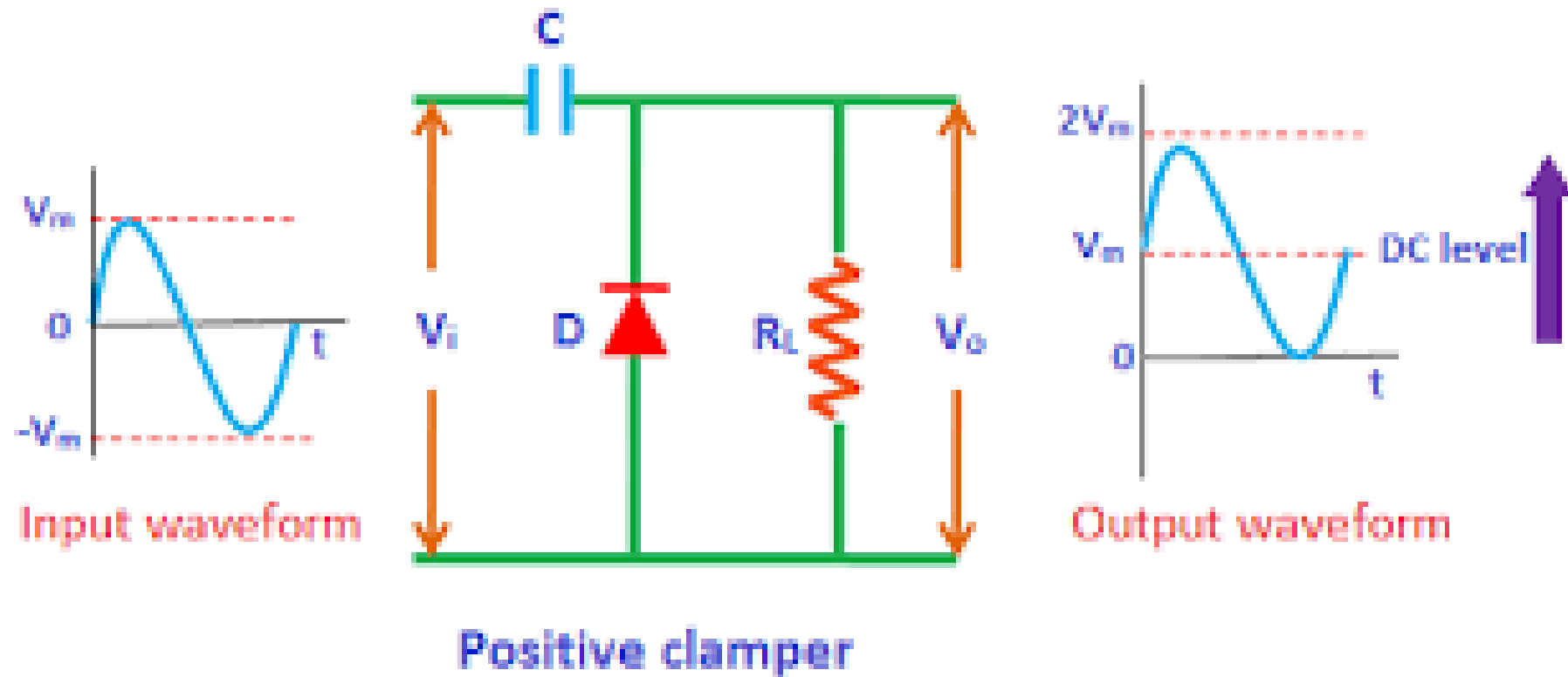


Series positive clipper

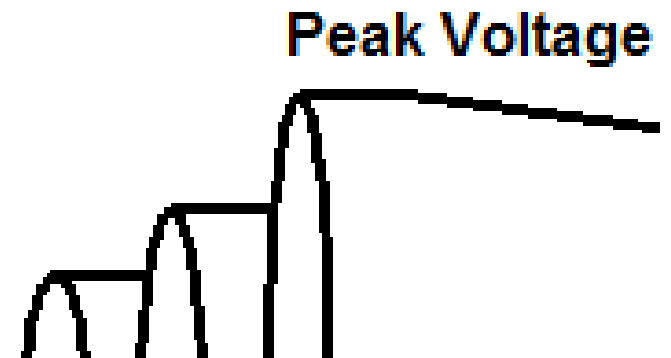
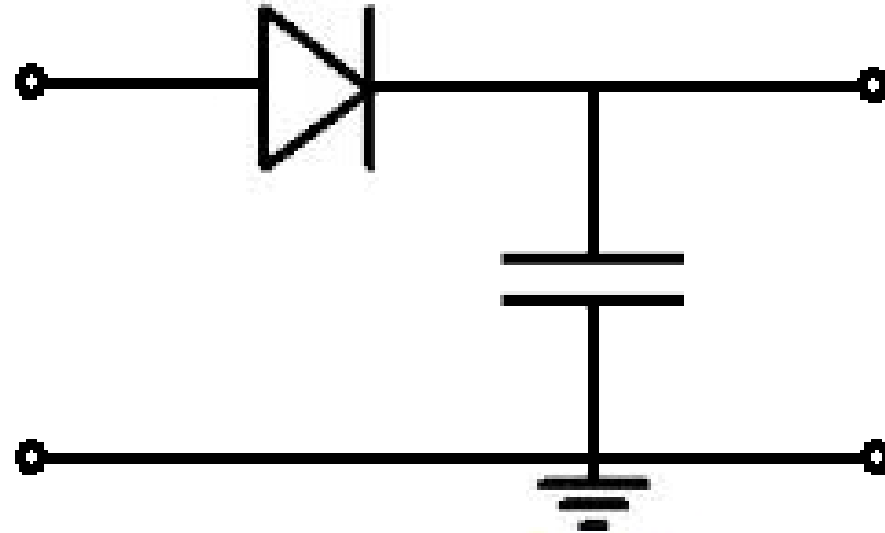
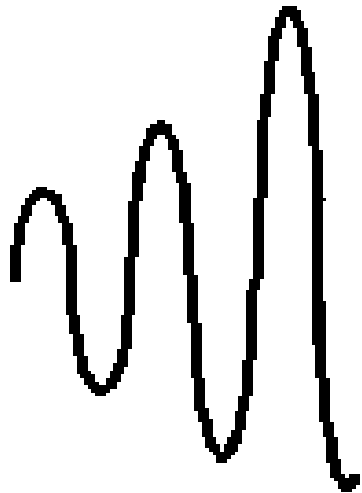
Application of Diode circuits (Clipping)



Application of Diode circuits (Clamping)



Application of Diode circuits (Peak detectors)



Application of Diode circuits (Rectification)

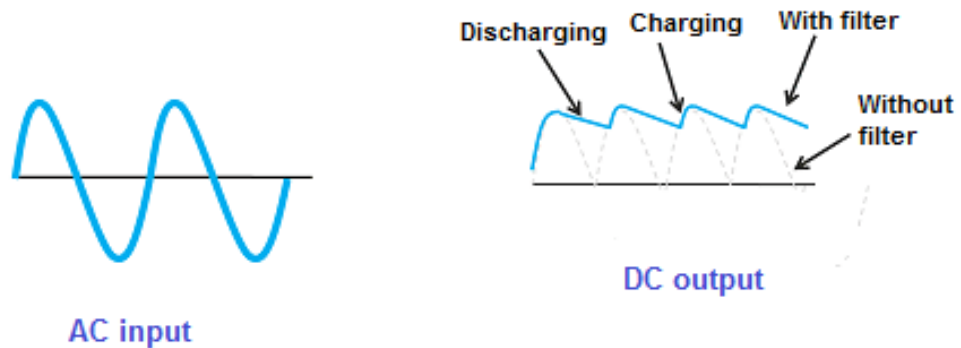
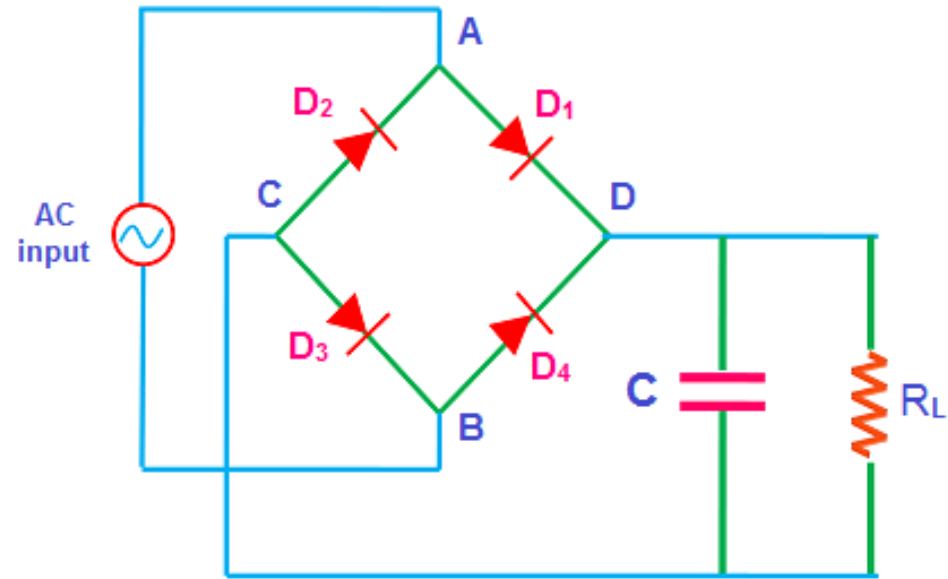
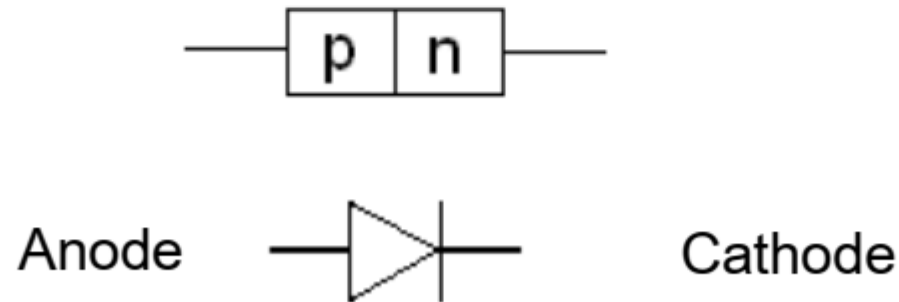


Fig: Bridge rectifier with filter

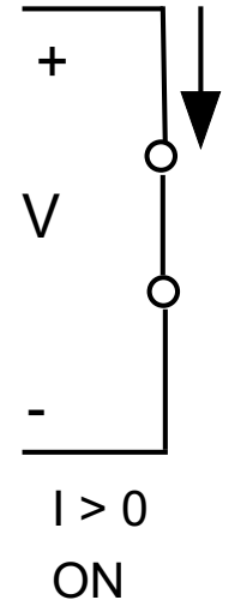
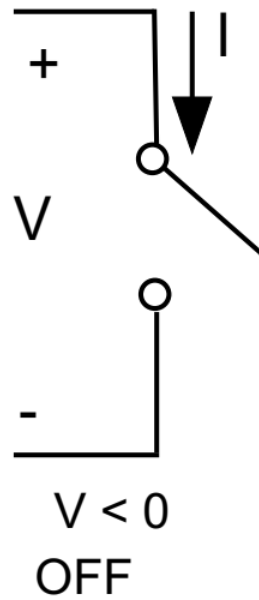
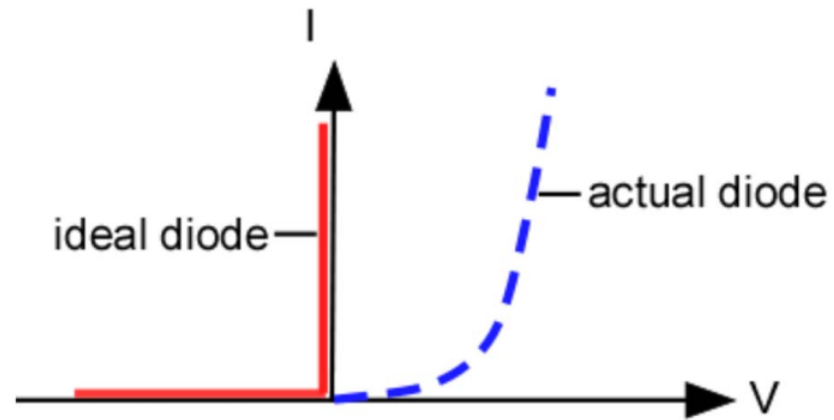
Semiconductor Diode

The semiconductor diode is formed by bringing p and n-type materials together (constructed from the same base – Ge or Si)

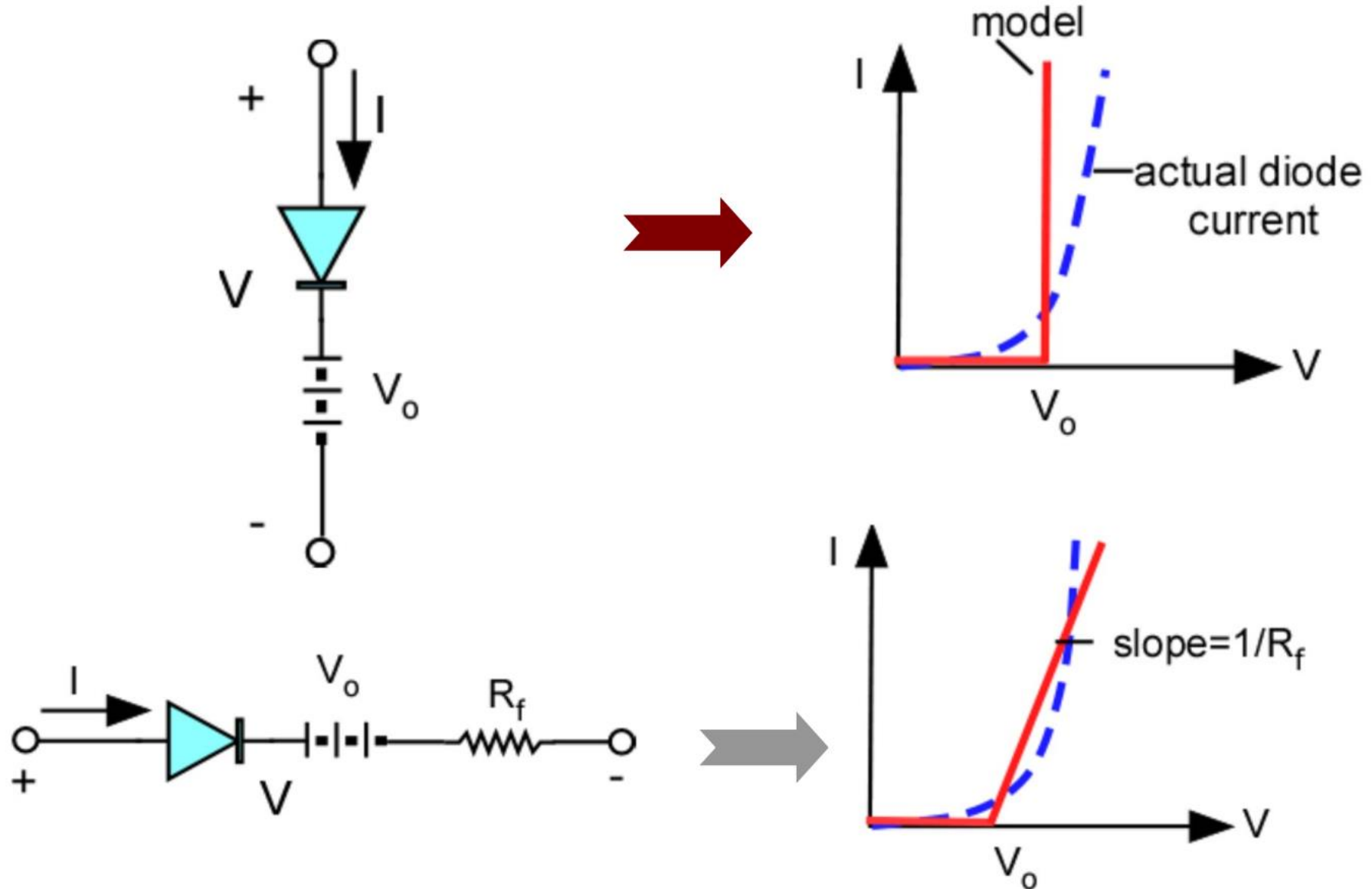


Diode Symbol

Ideal Diode Characteristics

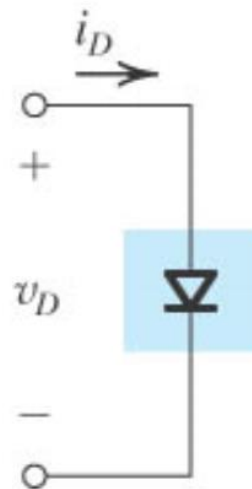
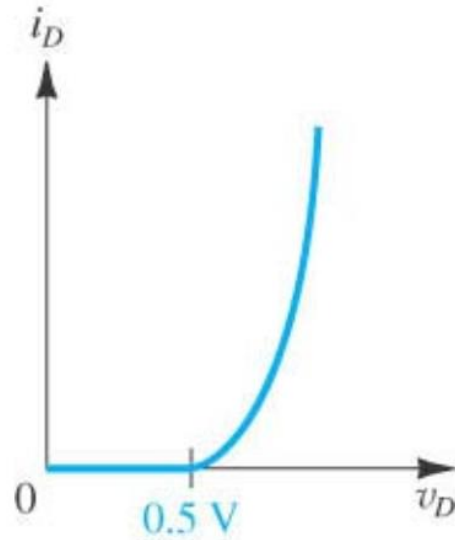


Ideal Diode Characteristics

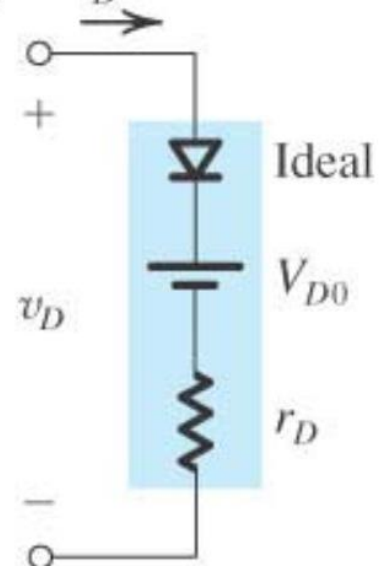
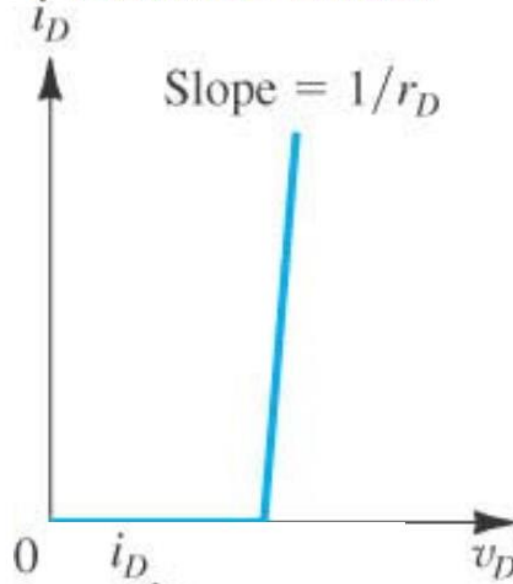


Diode Models

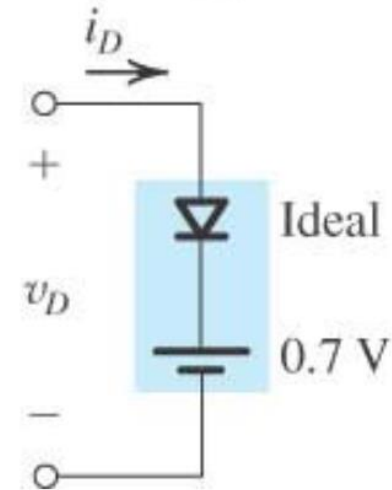
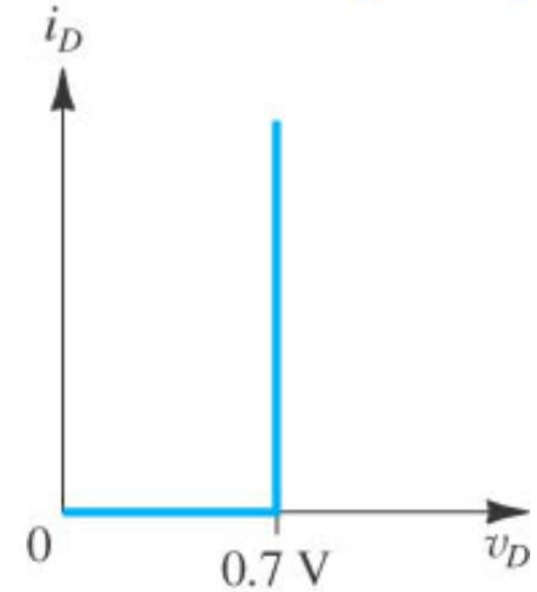
Exponential



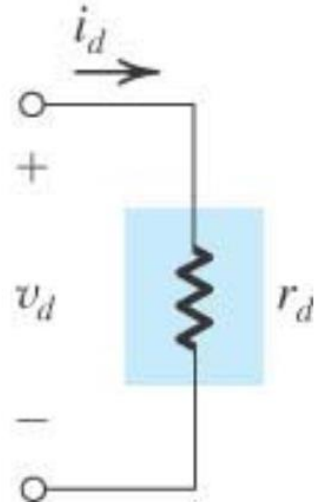
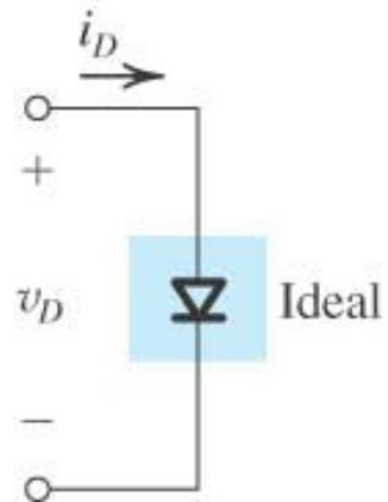
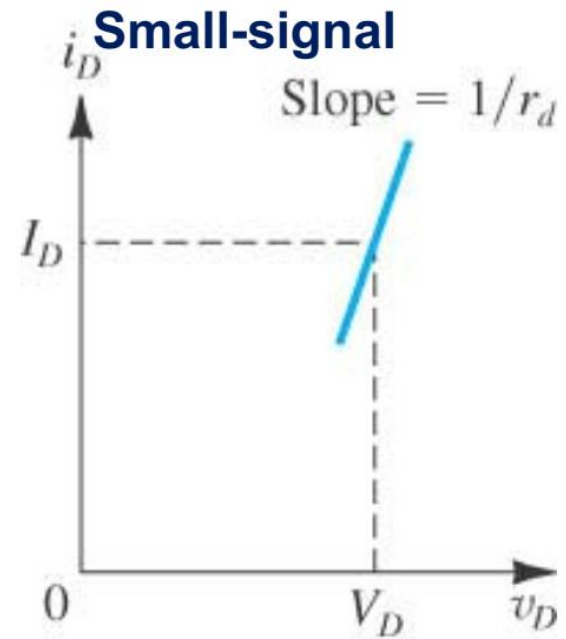
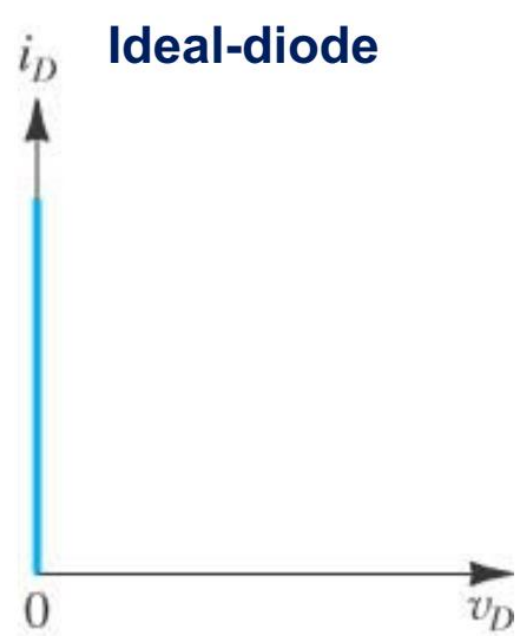
Piecewise Linear



Constant-Voltage-Drop



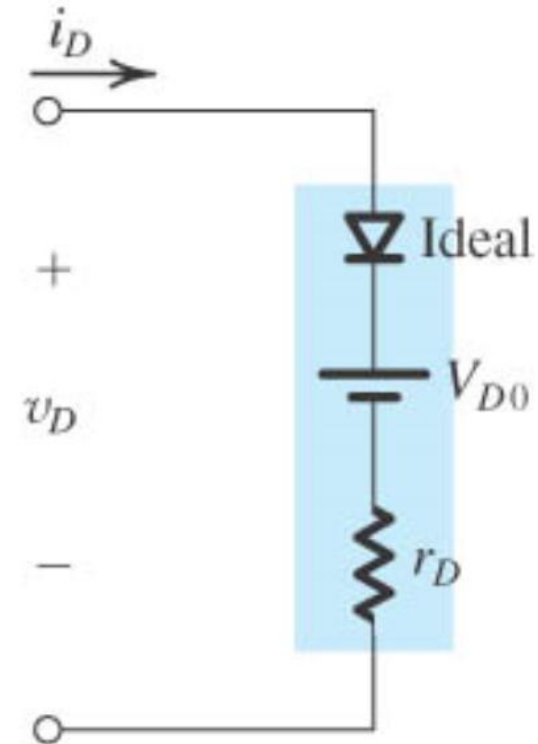
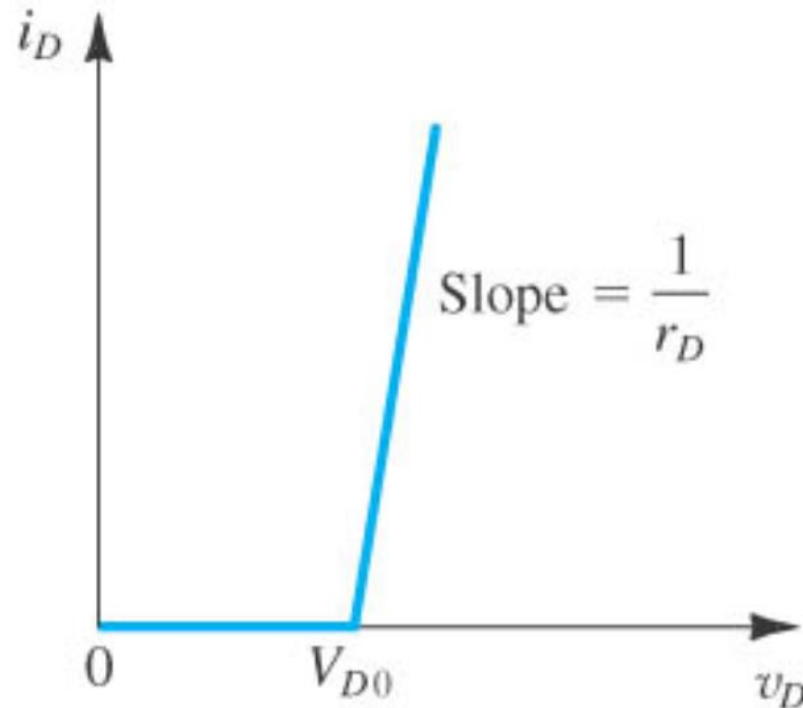
Diode Models



Piecewise-Linear Model

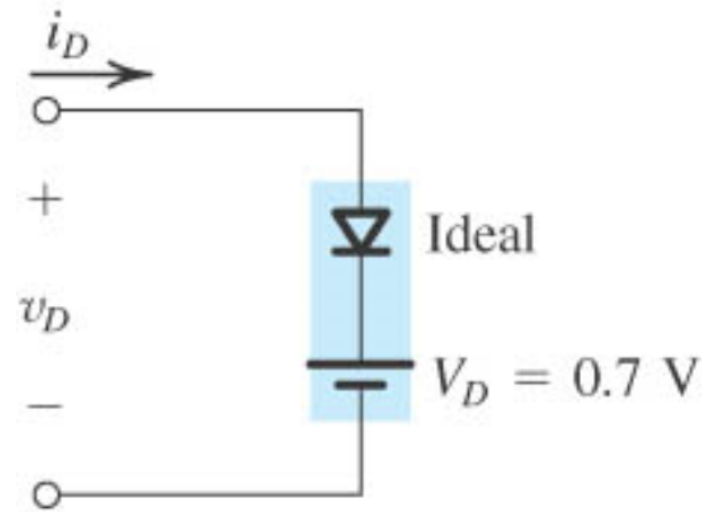
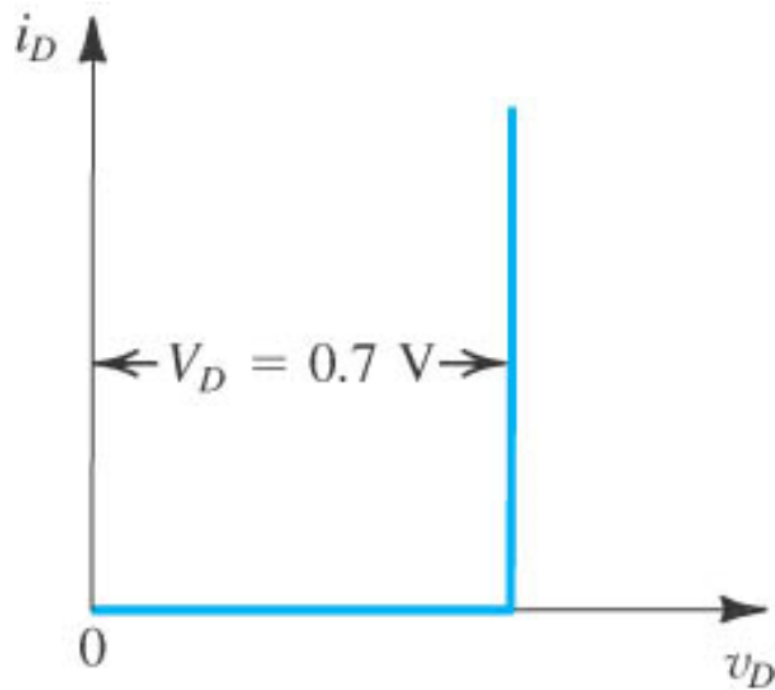
$$\text{for } v_D \leq V_{D0} : i_D = 0$$

$$\text{for } v_D \geq V_{D0} : i_D = \frac{1}{r_D}(v_D - V_{D0})$$



Constant-Voltage-Drop Model

for $i_D > 0$: $v_D = 0.7\text{ V}$



Diode Circuit Example 1

IDEAL Diodes

Assume both diodes are on; then

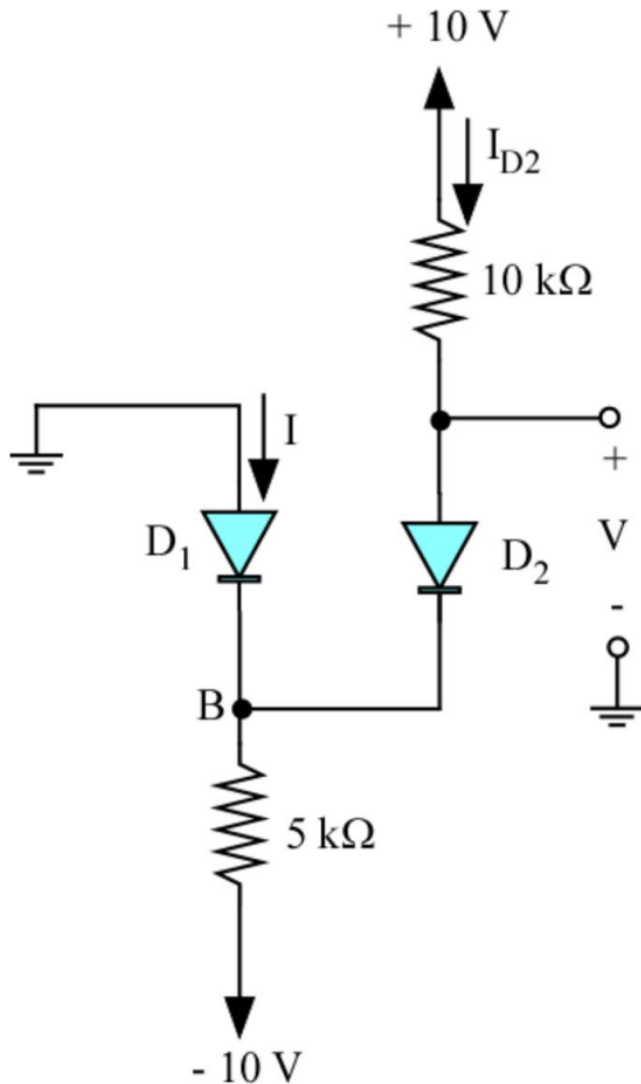
$$V_B = 0 \quad \text{and} \quad V = 0$$

$$I_{D2} = \frac{10 - 0}{10} = 1 \text{ mA}$$

At node B

$$I + 1 = \frac{0 - (-10)}{5} \Rightarrow I = 1 \text{ mA}, V = 0 \text{ V}$$

D₁ is conducting as originally assumed



Diode Circuit Example 2

IDEAL Diodes

Assume both diodes are on; then

$$V_B = 0 \quad \text{and} \quad V = 0$$

$$I_{D2} = \frac{10 - 0}{5} = 2 \text{ mA}$$

At node B

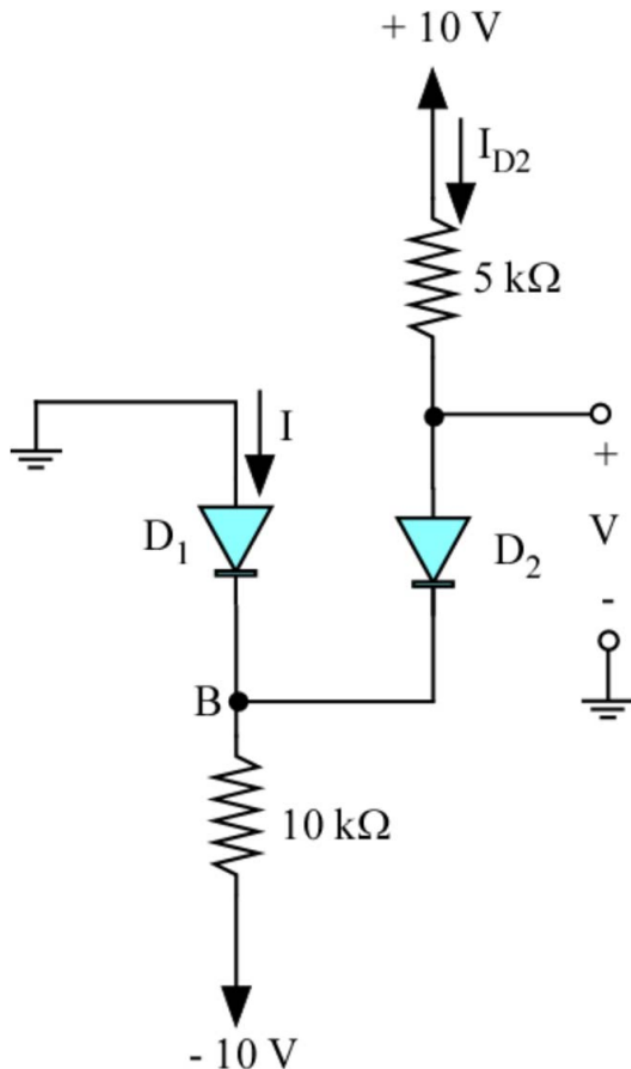
$$I + 2 = \frac{0 - (-10)}{10} \Rightarrow I = -1 \text{ mA} \Rightarrow \text{wrong}$$

**original assumption is not correct ...
assume D_1 is off and D_2 is on**

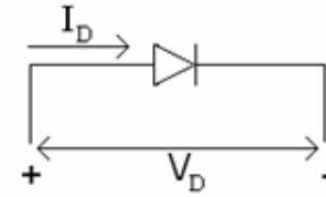
$$I_{D2} = \frac{10 - (-10)}{15} = 1.33 \text{ mA}$$

$$V_B = -10 + 10 \times 1.33 = +3.3 \text{ V}$$

D_1 is reverse biased as assumed



Diode equation



$$I_D = I_s (e^{V_D/nV_T} - 1)$$

where I_s is the reverse saturation current

V_D is the applied forward-bias voltage across the diode

n is an **ideality factor** which is a function of operating conditions and physical construction; it has a range between 1 & 2.

($n=1$ will be assumed unless otherwise noted)

$$V_T = \frac{kT}{q}$$

k is Boltzmann constant $= 1.38 \times 10^{-23}$ J/K

T is the absolute temperature in Kelvins

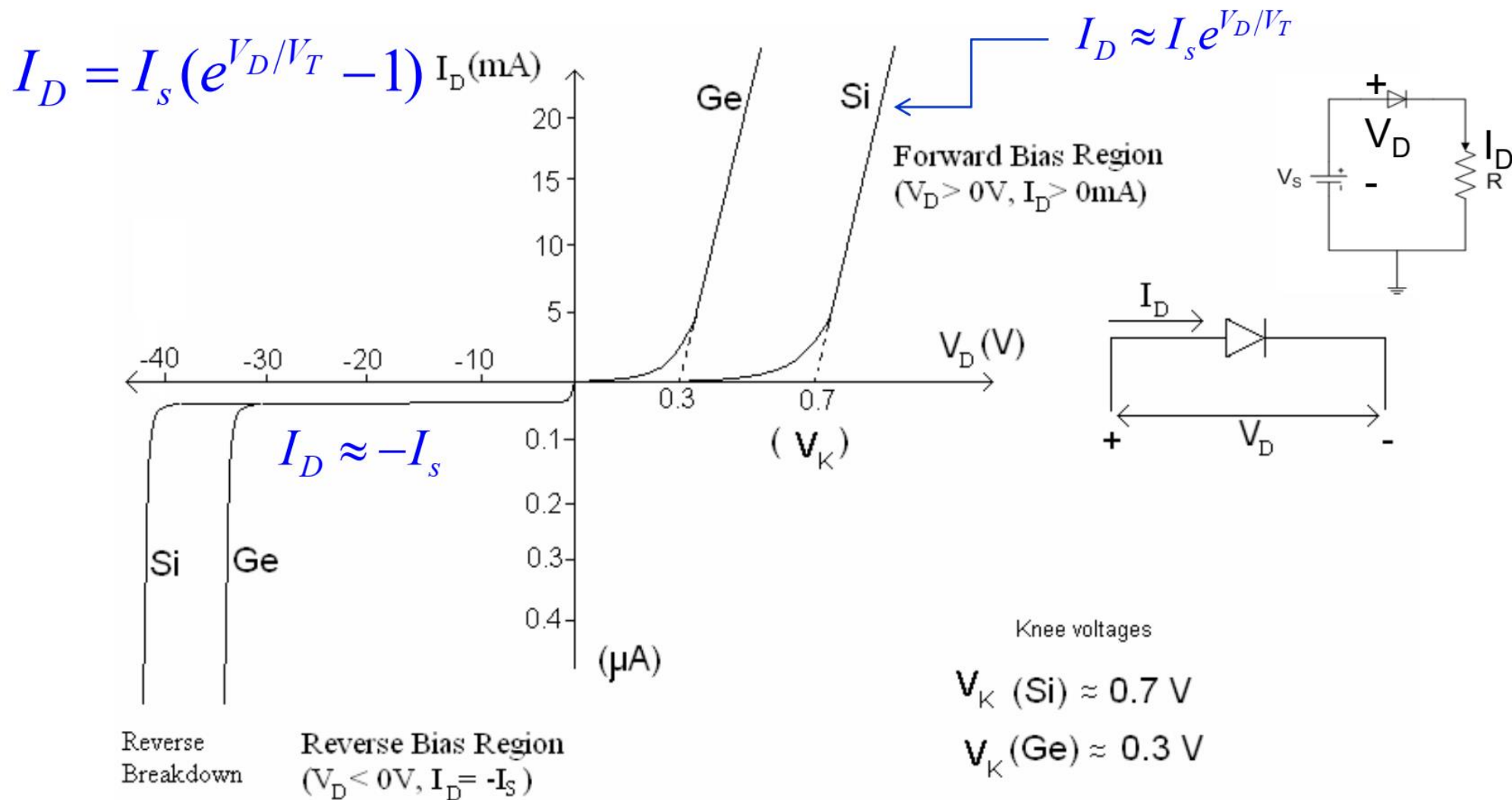
q is the electronic charge $= 1.6 \times 10^{-19}$ C

$V_T \approx 26$ mV at room temperature

For $V_D > 0$, $I_D \approx I_s e^{V_D/nV_T}$ and for $V_D < 0$, $I_D = -I_s$

Note that the scales on the different axes are different!

V-I characteristic of Diode



Temperature Effect

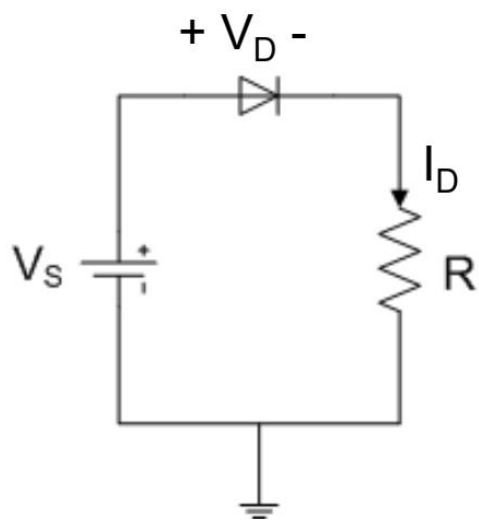
The reverse saturation current I_s approximately doubles for every 10°C rise in temperature.

If $I_s = I_{s1}$ at $T = T_1$, then at temperature T_2 , I_{s2} is given by,

$$I_{s2} = I_{s1} \times 2^{(T_2 - T_1)/10}$$

PIV (Peak Inverse Voltage Rating)

The maximum reverse-bias potential that can be applied to the diode without damaging it or causing it to break down.



$$I_D = \frac{V_S - V_D}{R}$$

$$I_D = \left(-\frac{1}{R}\right)V_D + \left(\frac{V_S}{R}\right)$$

